Lucas Leonel Galardo Euler

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> **Public version** — direct contact details (phone / email) are intentionally omitted to avoid bots/scrapers. Reach me via LinkedIn or Linktree.

Professional Summary

Video game and simulator programmer with **10+ years** of experience, specialized in **Unity (C#, URP, Burst, Jobs)** and the development of gameplay systems, internal tools, and editor plugins. Comfortable descending to **low-level** work (C++, DX11, Assembler). **Top of class (9.69/10)** in Virtual Simulator Design & Programming — Da Vinci. Experience in advanced technical pedagogy and leadership of multidisciplinary projects.

Professional Experience

Technical Consultant *(Part-time, partner track)* — Somos Leyenda

Remote · Spain | 2025 – Present
Stack: C#, Unity 6, URP, Burst, Jobs, ParrelSync, Photon Fusion 2.

- **Audited** Unity codebases produced by youth club members (architecture, scripts, networking, code quality) and delivered **canonical documentation** + P0–P3 improvement plans, surfacing structural bugs (static state not reset across scenes, race conditions).
- **Implemented** critical fixes and refactors when the audit flagged them as priority.
- **Led** the Somos Leyenda × Lifecole Game Jam (Mar 29, 2026): connected both organizations, **30 selected participants** out of a pool of **200–300 students**, **4 finalist teams** (winner: a student of mine). Designed full event logistics, role-balanced enrollment, pedagogical batch decomposition, and **paid one-on-one tutoring (€60 per 4.5h block)**. Prizes integrated into the **Allio Minecraft Launcher** of Somos Leyendas in production.
- **Currently undergoing partner-track promotion at the firm (2026)** — firm-level / equity partnership, not club membership.

Lead Instructor / Developer *(Part-time)* — Lifecole

Remote · Madrid, Spain | 2021 – Present
Sector: Online platform for technology after-school courses for ages 6–18, **3,400+ students trained**, live classes capped at 7 students per group.
Stack: Unity, C#, HLSL.

- **Designed the full *Unity Proficiency* curriculum** (the last 2 years of the Unity track, equivalent to the second half of the full learning path): quarterly with 3 axes (art, programming, design) × 4 levels, explicit pedagogical framework (measurable objectives, GitHub/Trello in the classroom), cross-stack pedagogy (programming + modeling + digital literacy + narrative). **100% own authorship.**
- **Lead instructor** for the two most advanced Unity levels (*Unity Advanced*, *Unity Proficiency*) and for the Minecraft (Joven, Expert) and Robotics tracks.
- **Delivered technical optimization talks** in the classroom (beyond curricular material).
- **Mentored** students up to competition and victory — **a student of mine won the Lifecole × Somos Leyenda Game Jam**.

Tech Lead / Product Engineer *(Part-time)* — Nexio

Remote · Burgos, Spain | 2026 – Present
Stack: TypeScript, React, Node.js, Supabase, PostgreSQL, Edge Functions, Google Gemini.

- **Led end-to-end** a multi-channel SaaS platform: roadmap, prioritization, backend architecture, product decisions, and cross-provider UX.
- **Designed and operated** the CI/CD pipeline (feature → staging → production) with versioned migrations and pull request reviews.
- **Integrated** third-party APIs and generative AI (Google Gemini / LLM integration).

Freelance Developer / Consultant

Remote | 2015 – Present

- **Mentored 40+ private students** (2021–Present) on Unity (games, university theses, capstones), Photon Fusion 2 multiplayer, and full-stack web. **Self-financed** my degree through this channel.
- **Built custom systems** for professional clients: sales, SQL-backed payments, mobile, crypto/Web3 backends, bots, Unity and Construct games.
- **Prototyped** 2D/3D video games with Unity (URP, Shader Graph, HLSL): mechanics, tools, and polish.

Software Developer *(Full-time)* — Wilson Champ S.R.L.

CABA, Argentina | 2015 – 2021
Stack: C, Assembler (Freescale), Arduino, PHP7, JavaScript, MySQL, Linux.

- **Developed** embedded software in C and Assembler for Freescale microcontrollers, with a focus on low-level optimization.
- **Built** IoT solutions with Arduino and ESP8266.
- **Led** networking and technical support projects.
- Parallel full-stack web development (PHP7, JS/jQuery/AJAX, HTML5, CSS3, MySQL).

Technical Skills

Languages: C# | C++ | C | HLSL | GLSL | Lua | Assembler | TypeScript | JavaScript | SQL | PHP

Engines and graphics: Unity 6 (URP 14→17, Shader Graph, Burst, Jobs, ScriptableRendererFeature, RTHandle, Depth Texture, UI Toolkit, ScriptableObjects, Assembly Definitions, Source Generators, AssetPostprocessor, SettingsProvider, DOTS/ECS, `Graphics.RenderMeshInstanced`, polymorphic `SerializeReference`, `UnityEngine.LowLevel.PlayerLoop`, Netcode for GameObjects, Photon Fusion 2) | DirectX 11 + Unity Native Plugins (C++↔C# interop, marshalling, x86/x64) | HLSL/GLSL | parallax occlusion | shadow atlas | OpenGL

Game programming: AI (hierarchical FSM with LCA, Steering Behaviours, GOAP) | pathfinding (Dijkstra/A*/Theta* over a custom binary heap with Decrease-Key) | Jobs+Burst-parallelizable spatial detection | hybrid `Parallel.For` + Jobs parallelization | GC-free data structures | unified motion systems (linear+angular) | multi-segment IK | procedural generation | Marching Cubes | modding API design (Lua for safe UGC)

Embedded / IoT: C and Assembler for Freescale | Arduino | ESP8266 | Raspberry Pi (local deployments / edge AI agents) | electronics

Web (full-stack background): React | TypeScript | Node.js | Supabase | PostgreSQL | REST APIs | applied prompt engineering

DevOps: Git | GitHub Actions | CI/CD | Linux | Visual Studio | Rider | cPanel/Zone Editor (DNS)

Methodologies: OOP | design patterns | dependency injection | concurrency and parallelism | **compile-time codegen (Source Generators + AssetPostprocessor)** as an IL2CPP-safe alternative to Reflection | automated unit testing (custom ScriptableObject-based architectures) | code review | technical pedagogy

Selected Projects

Common-Package — internal Unity library of reusable systems (2021 – Present)

10+ in-house unitypackages maintained over the years, with `.asmdef` per system; 100% own authorship. Technical highlights: - **Custom player loop** — extended phases (Pre/Post Update, EndUpdate) integrated via `UnityEngine.LowLevel.PlayerLoop` and recursive subsystem-tree mutation; sub-system auto-discovery via reflection over the `AppDomain`. - **GPU instancing** with `Graphics.RenderMeshInstanced` + parallel filling of `NativeArray` combining `Parallel.For` with `unsafe` pointers and Jobs+Burst per threshold. - **Hierarchical FSM with runtime Lowest Common Ancestor resolution** — invokes OnExit/OnEnter correctly when changing branches. - **Generic binary heap with O(log n) Decrease-Key** indexed by `Dictionary<T,int>` — brought Dijkstra/A _down from O(E log E) to O(E log V). - Lite DI with code generation — `AssetPostprocessor` + `[DidReloadScripts]` + reflection emits C# into `Generated/`. - Editor scaffolder meta-tool (an EditorWindow that generates other editor tools) and serializable `Type` selector over open generics with UI Toolkit. - MaterialPropertyBlockSetter — modify per-instance properties without breaking SRP batching or instantiating materials_*.

Unity library for HLSL compute-shader compilation on DX11 — Image Campus (Oct–Dec 2025)

Native C++ plugin that compiles **HLSL compute shaders at runtime** on DirectX 11 (x86+x64), integrated via `DllImport` and marshalling. **Automatic fragment→compute converter** with generated template (`RWTexture2D<float4>`); native DX compiler emits `.cso`; Unity loader via Reflection populates a `RenderTexture` at runtime. Solves Unity’s structural limitation (no runtime HLSL compilation through the standard API). **“Silent” GPU debugging (black screen with no log error) — 3 root causes resolved**: insufficient UAV caching, Main↔Render thread race, and **Constant Buffer 16-byte misalignment** silently zeroing the entire output.

Arrange the Heaven — Da Vinci thesis (2025)

Procedural RPG in Unity, team with Mouro and Revelli (role: programmer and designer). Combines procedural systems, modular AI (FSM/Steering/GOAP) and custom URP rendering. **Mass toggle of 4,096 GameObjects with hybrid Jobs + main-thread architecture** (Jobs for parallel state computation; `SetActive` on the main thread due to Unity API thread-safety). Showcases the reusable engine architectures I developed throughout the degree.

La Maldición CardGame — licensed digital adaptation (in active development, 2026)

Official digital adaptation of the Argentine card game *La Maldición*; secured the license and built the prototype. **Manual binary serialization subsystem on top of Netcode for GameObjects** (`INetworkSerializable`, `BufferSerializer<T>`, `RefList`) — replaced JSON in the multiplayer transport, achieving **significant bandwidth reduction**. O(1) indexing by `PlayerID`, zero allocations on the critical path. Multiplayer setup with ParrelSync. In active development; preparing a playable build for EVA (Argentina’s national game expo).

Education

Virtual Simulator Design & Programming — Da Vinci, CABA, Argentina (2021–2025)
Top of class · GPA: 9.69/10. Thesis: Arrange the Heaven (procedural RPG in Unity).

Shaders for Video Games / Computational Science — Image Campus (Oct–Dec 2025)
HLSL (Unity) and GLSL. Capstone: C++ library to compile HLSL compute shaders on DX11.

Electronic Engineering *(3 years completed)* — UTN, CABA, Argentina (2015–2018)
Electronics Technician — Colegio San José A-355 (2008–2014)
Complementary workshops at Da Vinci — HLSL, UX, sound for video games (2021–2025).

Languages

- **Spanish:** Native
- **English:** Professional Working Proficiency (advanced technical reading, professional writing, technical meetings)